

RECAP: For a time homogeneous DTMC $\{X_n, n \geq 0\}$ on the state space S .

- Transient, Null-recurrent & Positive Recurrent States
- Communicating classes & Class property theorem
- Irreducible DTMC, positive recurrence and a unique stationary distribution

PLAN: Examples

Irreducibility and Periodicity of a TH DTMC

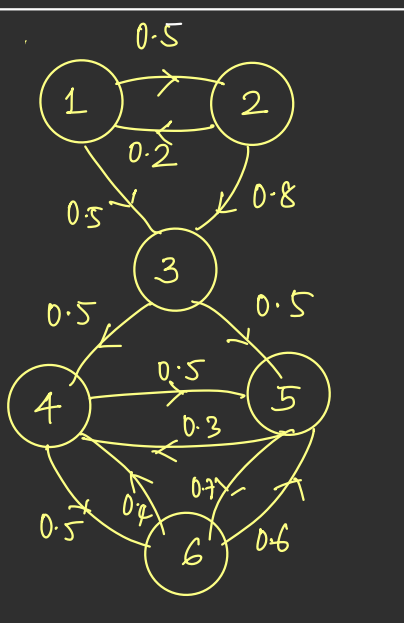
Proof of Class Property Theorem

Result about open/closed communicating classes

Homework: For j transient:

$$P(M_j = m | X_0 = j) = f_{jj}^m (1 - f_{jj})$$

Example: Consider the TH DTMC with the following transition probability diagram.



1) Transition Probability Matrix P

2) Assuming $\pi(0) = \text{unif } \{1, 2, 3, 4, 5, 6\}$, find $\pi(1), \pi(2)$

3) The first passage time distributions: $f_{ii}^{(n)}$, for all $i \in S$ and find f_{ii} , $\forall i \in \{1, 2, \dots, 6\}$.

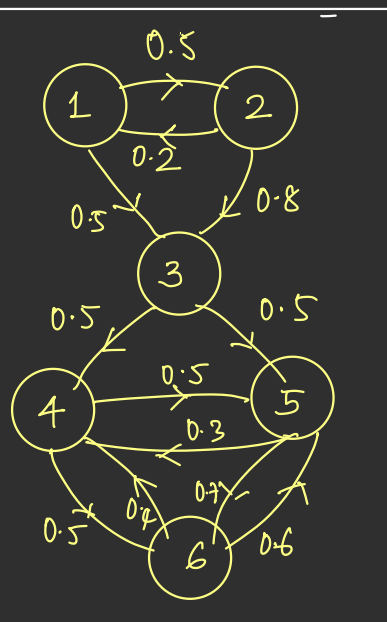
4) Classify the states as transient or recurrent based on the values of f_{ii} above

5) Reachability matrix and identify the communicating classes

6) If there are recurrent states, find their mean recurrence time v_{jj}

7) Based on v_{jj} , classify the recurrent states as positive or null recurrent.

8) Find the stationary distribution π s.t. $\pi = \pi P$.



$$1) P = \begin{bmatrix} 0 & 0.5 & 0.5 & 0 & 0 & 0 \\ 0.2 & 0 & 0.8 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.5 & 0.5 & 0 \\ 0 & 0 & 0 & 0 & 0.5 & 0.5 \\ 0 & 0 & 0 & 0.3 & 0 & 0.7 \\ 0 & 0 & 0 & 0.4 & 0.6 & 0 \end{bmatrix}$$

$$2) \pi(0) = \left[\frac{1}{6} \frac{1}{6} \frac{1}{6} \frac{1}{6} \frac{1}{6} \frac{1}{6} \right] \text{ Find } \pi(1) \text{ and } \pi(2)$$

$$\pi(1) = \pi(0) \cdot P$$

$$\pi(2) = \pi(1) \cdot P$$

$$3) f_{ii}^{(n)}: \quad f_{11}^{(n)} = \begin{cases} 0.1 & n=2 \\ 0 & \text{else} \end{cases} \quad f_{22}^{(n)} = \begin{cases} 0.1 & n=2 \\ 0 & \text{else} \end{cases}$$

$$f_{33}^{(n)} = \begin{cases} 0 & n=2 \\ 0.15 & n=3 \end{cases} \quad f_{44}^{(n)} = \begin{cases} 0.15 + 0.2 & n=2 \\ 0.14 + 0.09 & n=3 \\ \vdots & n=4 \end{cases}$$

$$f_{55}^{(n)}: \quad f_{66}^{(n)}:$$

5) Reachability Matrix

$$R = \begin{matrix} & j \\ i & \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix} \end{matrix}$$

Communicating classes.

$$C_1 = \{1, 2\}$$

$$C_2 = \{3\}$$

$$C_3 = \{4, 5, 6\}$$

Stationary distribution:

$$\pi = \pi P; \quad \pi_i \geq 0 \quad \forall i \in S \quad \text{and} \quad \sum \pi_i = 1$$

$$\begin{bmatrix} 0 & 0 & 0 & \pi_4 & \pi_5 & \pi_6 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & \pi_4 & \pi_5 & \pi_6 \end{bmatrix} P$$

$$\frac{1}{\sqrt{44}}$$

Perron Frobenius Theorem

If a matrix $A_{n \times n}$ is non-negative & it is irreducible

for every $i, j \in \{1, \dots, n\} \exists k \in \mathbb{N}$ for which $A_{ij}^{(k)} > 0$.

then $\exists$ a non-negative eigen vector corresp. to its spectral radius or 1st eigen value.

λ_{max}

TPM: P - row-stochastic matrix
DTMC is irreducible

$$\lambda_{max} = 1$$

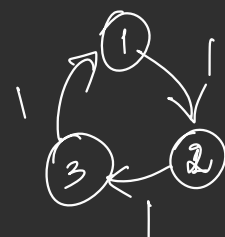
For an irreducible DTMC on a finite state space, a stationary distribution exists.

Periodicity of a DTMC. Consider a TH-DTMC $\{X_n, n \geq 0\}$ on a state space S with TPM P . For any $j \in S$ $R(j) = \{n \in \mathbb{N} \text{ s.t. } P_{jj}^{(n)} > 0\}$ denote the set of return times to j .

Defn: The period of a state $j \in S$, denoted $d(j)$ is the gcd of the return times, $d(j) = \gcd \{n \in \mathbb{N} \text{ s.t. } P_{jj}^{(n)} > 0\}$.

A state is aperiodic if $d(j) = 1$ else periodic.

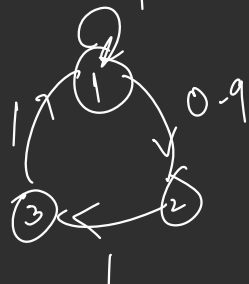
Example $P_1 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$



$$d(1) = d(2) = d(3) = 3.$$

→ Periodic DTMC

$$P_2 = \begin{bmatrix} 0.1 & 0.9 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$



$d(1) = 1$ 1 is an aperiodic state

$$d(2) = d(3) = 1$$

→ Aperiodic DTMC.

Aperiodic Markov Chain: A T.H-DTMC is called aperiodic if

i) it is irreducible

ii) every state is aperiodic.

Class Properties

1) Period is a class property, i.e. if $i \leftrightarrow j$, $d(i) = d(j)$.

Proof: $i \leftrightarrow j \Rightarrow \exists m, n \in \mathbb{N}$ s.t. $P_{ij}^m > 0$ and $P_{ji}^n > 0$.

Let $r \in R(i)$ i.e. $P_{ii}^r > 0$

$\uparrow$
set of all return
times to state i

$$\underbrace{P_{ji}^{m+n}} \geq \underbrace{P_{ji}^n} \cdot \underbrace{P_{ij}^m} > 0$$

$$\Rightarrow m+n \in R(j)$$

$$P_{jj}^{n+r+m} \geq \underbrace{P_{ji}^n}_{>0} \cdot \underbrace{P_{ii}^r}_{>0} \cdot \underbrace{P_{ij}^m}_{>0} > 0$$

$$n+r+m \in R(j)$$

$$d(j) \mid m+n \text{ and } d(j) \mid n+r+m \Rightarrow d(j) \mid r$$

$d(j)$ is a common divisor of $R(i)$

$d(j) \leq d(i)$ as $d(i)$ is the greatest common divisor of $R(i)$

To show: $d(i) \leq d(j)$

$$d(i) = d(j)$$

2) Transience and Recurrence are class properties

Recurrence: Suppose i is recurrent & $i \leftrightarrow j$ $\left. \begin{array}{l} \exists m, n \in \mathbb{N} \text{ s.t. } P_{ij}^m > 0 \\ P_{ji}^n > 0 \end{array} \right\}$

Recurrent: $\mathbb{E}[M_i | X_0 = i] = \sum_{k \in \mathbb{N}} P_{ii}^k = +\infty$

(expected # returns to state i)

$$\mathbb{E}[M_j | X_0 = j] = \sum_{k=1}^{\infty} p_{jj}^k \quad p_{jj} \geq p_{ji} \cdot p_{ii} \cdot p_{ij}$$

$$\geq \sum_{k=n+m+1}^{\infty} p_{jj}^k = \sum_{l=1}^{\infty} p_{jj}^{m+n+l} \geq \sum_{l=1}^{\infty} p_{ji}^n \cdot p_{ii}^l \cdot p_{ij}^m$$

$$\underbrace{p_{ji}^n \cdot p_{ij}^m}_{>0} \sum_{l=1}^{\infty} p_{ii}^l < \infty$$

Transience:

Suppose i is transient & $i \leftrightarrow j$

$\exists m, n \in \mathbb{N}$ s.t. $p_{ij}^m > 0$ and $p_{ji}^n > 0$

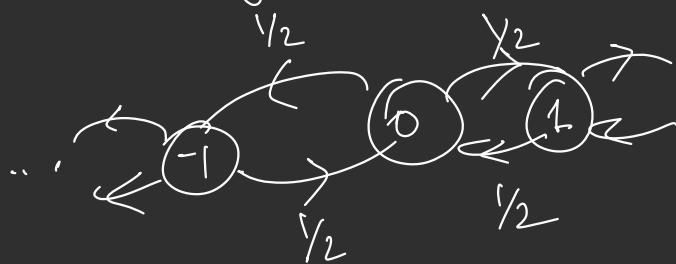
i is transient: $\mathbb{E}[M_i | X_0 = i] = \sum_{k \in \mathbb{N}} p_{ii}^k < \infty$

$$\mathbb{E}[M_j | X_0 = j] = \sum_{k \in \mathbb{N}} p_{jj}^k$$

$$= \frac{1}{p_{ij}^m \cdot p_{ji}^n} \sum_{k \in \mathbb{N}} p_{ij}^m p_{jj}^k p_{ji}^n$$

$$\leq \frac{1}{p_{ij}^m \cdot p_{ji}^n} \sum_{k \in \mathbb{N}} p_{ii}^{m+k+n} \leq \frac{1}{p_{ij}^m p_{ji}^n} \sum_{k \in \mathbb{N}} p_{ii}^k < \infty$$

Consider a simple symmetric random walk.



Null-recurrent

Irreducible DTMC

Consider state 0.

$$P(T_0 = 2) = (1/4 + 1/4)$$

$$P(T_0 = 2n)$$

5) Null recurrence & positive recurrence are also class properties.

Results: 1) C is any open communicating class, then all states within C are transient

2) C is any finite closed communicating class, then all states are positive recurrent.
